Zeyad Mahmoud Manaa

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RESEARCH INTERESTS

Dynamical systems, reduced-order modeling, control theory, machine learning, and optimization.

EDUCATION

- **King Fahd University for Petroleum and Minerals** **Dhahran, Saudi Arabia**
  M.Sc in Aerospace Engineering *January 2023 – 2025 (expected)*
  Current GPA: 4.00/4.00
  Research on data-driven modeling and control for dynamical systems involving Koopman operator theory, physics informed machine learning, and sparse identification of nonlinear dynamics.

- **University of Science and Technology at Zewail City** **Giza, Egypt**
  B.Sc in Aerospace Engineering *September 2017 – June 2022*
  Thesis: Development of the software package for the attitude determination and control algorithm of a cube satellite.

RELEVANT RESEARCH EXPERIENCE

- **Research Assistant** **Dhahran, Saudi Arabia**
  King Fahd University for Petroleum and Minerals - Flight Dynamics and Control Lab *January 2023 - Present*
  Supervisor: Prof. Ayman M. Abdalla
  - Conducting research on a data-driven discovery of the quadrotor's equations of motion using Sparse Identification of Nonlinear Dynamics.
  - Research on Koopman operator modeling to globally linearize nonlinear dynamics and its intersection with deep learning in aerospace applications.
  - Worked on the basis of reduced order modeling for high dimensional systems using proper orthogonal decomposition and dynamic mode decomposition.
- **Research Intern** **Islamabad, Pakistan**
  National University of Science and Technology - Aerial Robotics Lab *June 2022 - August 2022*
  Supervisors: Prof. Muhammad Jawad Khan and Muhammad Aleem Siddiqui
  - Developed an autonomous control system for a quad-copter drone (Tello drone).
  - Simulated the control algorithm using **ROS**. (Worked on COEX Clover 4.2).
  - Used hand gestures to control the drone via built-in/web camera using **ML/AI** (Tello drone) – Done using **socket programming** (Server/Client).
- **Bachelor's Thesis Intern** **Giza, Egypt**
  ADCS Lab at Egyptian Space Agency – University of Science and Technology at Zewail City *August 2021 - July 2022*
  Supervisor: Prof. Mustafa Abdalla
  - Developed the software of the **attitude determination and control** algorithm of a cube satellite.
  - Modeled the space environment as means of Earth's Magnetic Field (IGRF Model), Earth's gravitational Field (Spherical Harmonics) as well as modeling the space disturbances using **MATLAB**.
  - Implemented the control algorithms (**$\dot{B}$, LQR, and PD**) for satellite's modes of operation.
  - Developed an efficient control scheme based on magnetic actuators in terms of maneuver time, robustness, stability, and applicability for implementation on an on-board computer.

PUBLICATIONS

1. **Z. M. Manaa**, M. R. Elbalshy, and A. M. Abdallah, *Data-driven Discovery of The Quadrotor Equations of Motion Via Sparse Identification of Nonlinear Dynamics.* arXiv, May 25, 2023. doi: 10.48550/arXiv.2305.16500. (Accepted for AIAA SciTech Conference 2024)
2. **Z. M. Manaa**, N. Qasem, *Novel airfoil for enhanced supersonic aerodynamic performance using convex optimization approach.* (Submitted to J. of Aerospace Science and Technology)

TEACHING

- **Instructor** — **Dhahran, Saudi Arabia**
  King Fahd University for Petroleum and Minerals — *2023 - present*
  - **Courses**: AE 426 - Flight dynamics and control
  - **Labs**: AE 315 - Control systems

SKILLS

Programming Languages:
MATLAB/Simulink (4 years):
Used MATLAB to model dynamical systems, earth's magnetic field, earth's gravity model, and to implement simulation environments to test control algorithms.
C/C++ – Object Oriented(2 years):
Used C++ to implement control algorithms used in hardware.
Python (Sci-kit learning, TensorFlow) (2 year):
Used python to develop ML models, simulate dynamical systems, and test control algorithms.

Software:
SolidWorks (2 years):
Used SolidWorks to model mechanical components such that a reaction wheel, gear box components.
ANSYS (2 year):
Used to run CFD simulations for airfoil optimization purposes alongside with convex optimization
Other Skills & Technologies:
- Familiar with Linux, GitHub, Mathematica, Bash-scripting, and LabView.
- Have a good knowledge regarding communication theory.

SELECTED ACADEMIC PROJECTS

- **Data-driven Approximation of Koopman Operator** — *June – 2023*
  - Finding global linear model for twin rotor solely from data using Koopman operator approximation via Extended Dynamic Mode Decomposition.
- **On Proper Orthogonal Decomposition and Dynamic Mode Decomposition for Aerodynamics** — *April – 2023*
  - Explore the potential of these techniques in analyzing large and complex data resulting from complex dynamical systems, especially in aerodynamics using model reduction techniques.
  - Potential system identification using Dynamic Mode Decomposition which further promote future state prediction.
- **Quadcopter Control Using Hand Gestures** — *June – 2022*
  - Controlling drone functionalities using hand gestures via the laptop webcam using **ML**.
  - Used socket programming (**Server/client programming**) to enhance the speed of the operations.
- **Attitude Determination & Control of CubeSat** — *July – 2021*
  - Development of the Attitude Determination and Control Subsystem for one degree of freedom in a CubeSat.

AWARDS AND ACHIEVEMENTS

- **Graduate International Research Assistance Scholarship – 2023**
  - *King Fahd University for Petroleum and Minerals, Aerospace Engineering department.*
- **Fully Funded Internship Program – 2022**
  - *Offered a fully-funded Internship program to conduct research at the National University of Science and Technology, Islamabad, Pakistan.*
- **Future Work is Digital Scholarship – 2021**
  - *Offered a fully-funded scholarship from the ministry of communications and information technology to study **data analysis from Udacity nano-degree**.*
- **Smart City Hackathon – 2019**
  - *Global finalist, awarded the **1st place** award for designing mobile application serving medical help in smart cities.*
- **Full Undergraduate Scholarship – 2017**
  - *University of Science and Technology at Zewail City.*

VOLUNTARY ACTIVITIES AND OTHER SKILLS

- **Leadership:** Led a group of 10 people in ***Euroavia Egypt*** while working on the media campaign. Furthermore, led a group of 15 people in ***Zewail City Science Festival*** mega event.
- **Cultural competence:** Worked with teams from different countries, such as France, United Kingdom, USA, Spain, and Italy.
- **Languages:**
  - Arabic (Fluent).
  - Have a full command of English with a professional working proficiency with an overall **IELTS score of 7.0/9.0 – C1** User.